Anh Tuan LE (ORCID 0000-0003-1596-5465)

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Laboratoire de Mécanique des Solides – LMS
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ACADEMIC BACKGROUND

2017-2021	Doctorate in Mechanics and Materials Science, Institut Clément Ader, École des Mines Thesis published in *theses.fr* (DOI)
2015-2017	Master degree in Materials and Structures for Sustainable Construction Université Claude Bernard Lyon 1
2012-2015	Bachelor degree in Mechanics, Civil Engineering Université Claude Bernard Lyon 1
2012	French Baccalaureate & Vietnamese Baccalaureate High school for the gifted Ly Tu Trong, Can Tho, Vietnam

RESEARCH EXPERIENCE

2022-2025	Postdoctoral researcher, Laboratoire de Mécanique des Solides, École polytechnique, France *Subject:* Effect of stress with biaxiality ratios on the electrochemical responses of silicon electrode
2017-2021	Doctorate, Institut Clément Ader, École des Mines *Subject:* Manufacturing and mechanical characterisation of Vertically Aligned Carbon NanoTubes enhanced continuous fibre reinforced polymer composites (DOI)
2017	Research internship, Master 2, Université Claude Bernard Lyon 1 *Subject:* Influence of curing conditions on the mechanical behaviour of adhesive joints of carbon fibre-reinforced polymer/cement composites

RESEARCH AREAS

Electro-chemo-mechanical coupling;
Silicon-based lithium-ion batteries;
Manufacturing of composite materials (autoclave and vacuum bagging);
Mechanical characterisations of materials;
Thermoset matrix epoxy: mechanical and physical properties.

MACHINE LEARNING / DEEP LEARNING

Machine Learning Techniques in MATLAB (certificate);
Deep Learning Techniques in MATLAB for Image Applications (certificate).

LANGUAGES

English	●●●●●○	C1 Advanced – Cambridge English – Pass
Français	●●●●●●	Bilingual
Tiếng Việt	●●●●●●	Mother tongue

REFEREES

Professor emeritus **Michel ROSSO**
Laboratory of Condensed Matter Physics
École polytechnique
CNRS research director
michel.rosso@polytechnique.edu

Professor **François OZANAM**
Laboratory of Condensed Matter Physics
École polytechnique
CNRS research director
francois.ozanam@polytechnique.edu

Assistant professor **Laurent GUIN**
Laboratory of Solid Mechanics
École polytechnique
laurent.guin@polytechnique.edu

TEACHING EXPERIENCE

From 2022 — **École polytechnique & Polytech Sorbonne**
32 hours of practical sessions, Advanced Lab, second-year bachelor's students, École polytechnique
64+32 hours of practical sessions, Discovery Lab, first-year bachelor's students, École polytechnique
16 hours of tutorials, engineering students, Structural Vibrations, Polytech Sorbonne

2017-2021 — **École des Mines**
40 hours of tutorials, engineering students, Strength of materials, Additive manufacturing & Structural properties of materials, IMT Mines Albi

2016-2017 — **Université Lyon 1**
62 hours - Monitor in Physics

CROSS-DISCIPLINARY EXPERIENCE

2017-2023 Student expert – High Council of Evaluation of Research and Higher Education (HCERES):
Department of evaluation of programmes: CNAM Paris (Bachelor's and Master's programmes – 2017), Université Bretagne Sud (Masters – 2021) ;
Department of evaluation of institutions: École Supérieure du Bois (2021) ;
Department of evaluation of doctoral programmes: Aix-Marseille Université (2022).

2017-2021 Scientific popularisation: European Researcher's Night (city of Albi - 2018), Three minutes thesis (Toulouse - qualified – 2020).

2017-2019 Representative of PhD students:
Research committee of Ecole des Mines (2018-2019) ;
Institut Clément Ader Laboratory Council (2018-2019).

COMMUNICATIONS

2025 *Effect of Stress with Biaxiality Ratios on the Electrochemical Response of Amorphous Silicon Electrodes*
Oral presentation – 12th European Solid Mechanics Conference 2025, Lyon, France

2023 *Effect of Stress on the Electrochemical Response of Amorphous Silicon Electrodes*
Oral presentation – 243rd ECS Meeting, Boston, USA – DOI
Oral presentation - 12th Battery Day in Île-de-France

2019 *Impregnation and transfer of vertically aligned carbon nanotubes forests onto prepreg*
21è Journées Nationales sur les Composites – HAL Open Archive

PUBLICATIONS

2025 *Stress dependence of the chemical potential of lithium in a silicon electrode*
LE *et al.*, Journal of the Mechanics and Physics of Solids, The Nicolas Triantafyllidis 70th Anniversary Volume 207, February 2026, 106421 – HAL Open Archive – DOI

2023 *Composite laminates reinforced by vertically aligned carbon nanotubes: Detailed manufacturing process, from nanotubes transfer to composite consolidation*
LE *et al.*, Journal of Composite Materials, Vol. 57, Issue 27, p. 4239–4253 – HAL Open Archive – DOI

2023 *Mode I and mode II fracture behavior in nano-engineered long fiber reinforced composites*
LE *et al.*, Polymer Composites, Vol. 44, Issue 7, p. 4016-4026 – HAL Open Archive – DOI

2023 *Nano-engineered prepreg manufacturing: control of capillary rise of resin into VACNTs' forests*
LE *et al.*, Carbon Letters, Volume 33, p. 1019-1025 – HAL Open Archive – DOI

2019 *Influence of curing conditions on mechanical behaviour of glued joints of carbon fibre-reinforced polymer composite/concrete*
LE *et al.*, Construction and Building Materials, Vol. 227 (2019) 116385 – DOI